

Valuation of Options - part 1

Quantitative Finance

Tilburg University

Ramon van den Akker

Section 1

Introduction

Options (quick recap)

Call option

- contract between buyer (option holder) and seller (option writer)
- contract offers buyer *right* (but *not obligation!*) to *buy* specified security (or other *underlying financial asset*) at *specific* date (*maturity/expiration date*) or during a specified period of time at *agreed upon price* (*strike price*)

Put option

- analogous definition; but put offers buyer right to *sell*

Remarks

- As a put or call provides its buyer a right and not an obligation, the contract has a (positive) price.
- Please note that the seller (writer) of the contract has the obligation to sell (call) or buy (put) in case the option is exercised by the buyer.

Options (quick recap)

Underlying asset:

- stock, index, ETF, interest rate(s), real estate, etc.

Exercising the option:

- *European (style) options*: option can only be exercised *at expiration date*
- *American*: option can be exercised *at any time between purchase and expiration date*
- (there also exist *Asian*, *Bermudan* and *Canary* versions; all these names have nothing to do with geographical locations)
- typically: options on indices and interest rates are European style, while options on (individual) stocks and ETFs are American style

Further terminology:

- *strike price* is a.k.a. *exercise price*
- final date for exercising option: *maturity* or *expiration date*

Remarks:

- options are traded on exchanges, in Over-The-Counter markets, and in bespoke contracts (investment banks)
- this course focuses on European style options (on stocks, indices, ETFs); see MSc QFAS for American style options and options on interest rates
- options are examples of *financial derivatives* which are contracts that are defined in terms of underlying asset which is traded on financial market

Options (quick recap)

Exercise

Check the following formulas for the payoff at maturity T (we denote a generic payoff by C_T).

- European call option with strike price K : payoff $\max\{S_T - K, 0\}$
- European put option with strike price K : payoff $\max\{K - S_T, 0\}$

We will often use the following options that are defined by their pay-off:

- European digital (binary) call option with strike price K : payoff $1\{S_T \geq K\}$
- European digital (binary) put option with strike price K : payoff $1\{S_T \leq K\}$

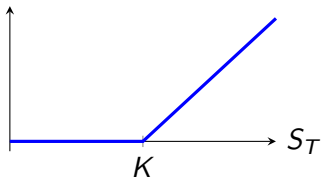
Please recall that the indicator $1\{A\}$ equals 1 if statement A is true and 0 otherwise.

Note that we can write each of these payoffs as a function of S_T , i.e. $C_T \equiv \sigma(S_T)$ for some function σ .

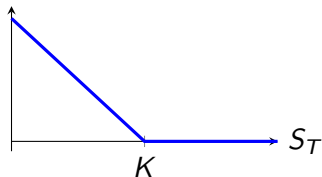
Payoff diagrams at maturity T

The payoff $C_T = g(S_T)$ of each contract, as a function of the stock price S_T :

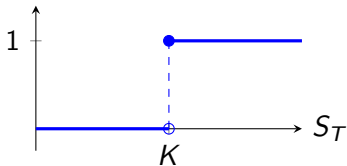
call: $\max\{S_T - K, 0\}$



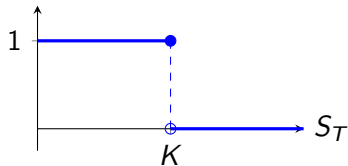
put: $\max\{K - S_T, 0\}$



digital call: $1\{S_T \geq K\}$

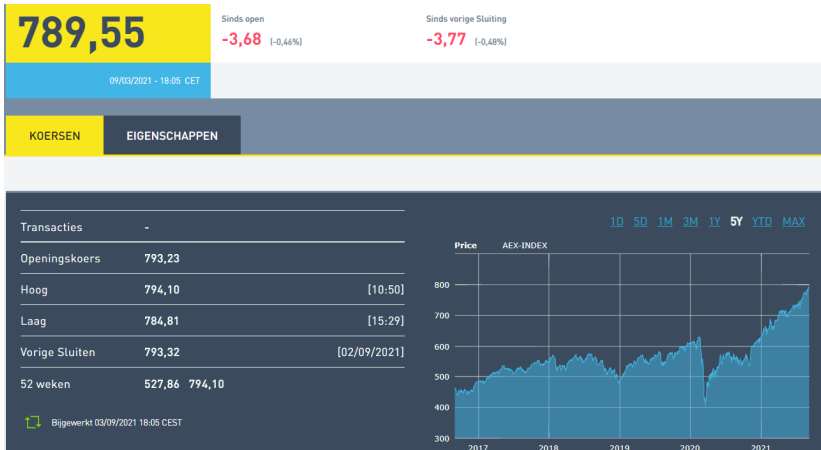


digital put: $1\{S_T \leq K\}$



Options (quick recap)

As an example, we will consider European call and put options on the AEX-index at September 3, 2021. (Illustrative market data; the precise figures are not important.)



Options (quick recap)

As an example, we will consider European call and put options, on the AEX-index at September 3, 2021, with expiration date in December 2022.

AEX-INDEX

AEX

Aantal 03/09/2021 17:29

24 732

Openstaande positie 02/09/2021

287 683

Valuta

EUR

Tijdzone

CET

DECEMBER 2022 KOERSEN - 03/09/21

	SETTL	LAATSTE	BIED	LAAT		UITOEFENPRIJS		BIED	LAATSTE	SETTL
+	230.19	-	225.25	238.95	C	550.00	P	13.60	14.10	14.03
+	185.35	185.00	183.00	188.50	C	600.00	P	19.10	18.90	19.56
+	142.83	-	141.00	144.00	C	650.00	P	26.80	27.20	27.40
+	103.35	100.00	102.00	104.00	C	700.00	P	37.50	38.20	38.29
+	68.30	68.20	67.60	68.90	C	750.00	P	52.70	-	53.62
+	40.26	41.55	39.50	40.70	C	800.00	P	74.10	76.80	75.94
+	20.67	20.45	20.30	20.90	C	850.00	P	-	106.50	106.72
+	9.39	9.50	9.05	9.65	C	900.00	P	140.00	-	145.81
+	4.16	3.90	3.85	4.40	C	950.00	P	-	-	190.95
+	1.89	2.00	1.70	2.15	C	1000.00	P	-	-	239.05
+	0.53	0.63	0.25	0.80	C	1100.00	P	-	-	338.42

Valuation/pricing

What is the 'fair' price of an option and how does it evolve over time? This is an important question for:

- investors/traders
 - to identify that you don't pay too much
 - to make money in case you think the market price is wrong
- financial institutions
 - who need to calculate price for selling OTC-options (or to price 'embedded' options)
 - who need to perform risk management on portfolios that contain options

Remark: later on we will also discuss **hedging**.

Main problems - valuation

- we are interested in (fair) price C_t , for $t \in [0, T)$, of (European) option with payoff $C_T = f(T, S_T)$ at maturity T
- we will learn three methods to determine 'fair price' (using specified continuous-time model for financial market):
 - Black-Scholes Partial Differential Equation
 - risk-neutral pricing
 - pricing kernel
- in particular: you will derive the famous Black-Scholes price for European put option with maturity T and strike K :

$$C_t = p(S_t, T - t, K, r, \sigma) = K e^{-r(T-t)} \Phi(-d_2) - S_t \Phi(-d_1)$$

$$d_1 = \frac{\log\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t},$$

- does not depend on μ !

Black-Scholes formula



- in 1973 Fisher Black and Myron Scholes published paper 'The Pricing of Options and Corporate Liabilities' and Robert Merton published 'Theory of Rational Option Pricing'
- Merton and Scholes received Nobel Memorial Prize in Economic Sciences in 1997 (Black died in 1995)

All pricing methods we discuss rely on insisting that arbitrage opportunities do/should not exist.

Definition

If it is possible to find an admissible self-financing trading strategy with value process V such that, for some $T > 0$, we have $V_0 \leq 0$ and

$$\mathbb{P}(V_T < 0) = 0 \text{ and } \mathbb{P}(V_T > 0) > 0,$$

then we say that an arbitrage opportunity exists.

Remarks:

- in QF you are allowed to ignore 'admissible': this condition rules out 'doubling strategies'. A sufficient condition for admissibility: $\mathbb{E}V_t^2 < \infty$ for all $t \in [0, T]$, where V_t denotes the portfolio value of the self-financing strategy

Section 2

Toy model

First we discuss a very simple discrete time model and discuss pricing of options under assumption that arbitrage opportunities do not exist. We discuss two methods:

- pricing by replication
- risk-neutral pricing

Later on we will discuss their continuous-time analogues. Although the math gets more complicated, the underlying ideas are the same!

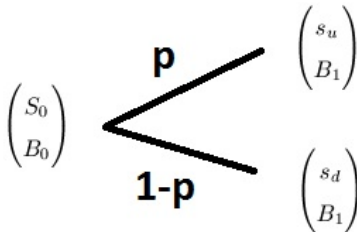
Model for financial market

- two states-of-the-world: $\Omega = \{\text{"up"}, \text{"down"}\}$
- two assets: risky asset S and riskless asset B
 - $B_1(\text{"up"}) = B_1(\text{"down"}) = B_1 > 0$
 - for

$$S_1(\omega) = \begin{cases} s_u, & \text{if } \omega = \text{"up"} \\ s_d, & \text{if } \omega = \text{"down"} \end{cases}$$

we assume $0 < s_d < s_u$

- probability of state "up" is $p \in (0, 1)$, i.e. $\mathbb{P}(\text{"up"}) = p$



Theorem There are no arbitrage opportunities in our financial market if and only if

$$0 < \frac{s_d}{S_0} < \frac{B_1}{B_0} < \frac{s_u}{S_0}.$$

Proof

' $\Rightarrow$ ': exercise

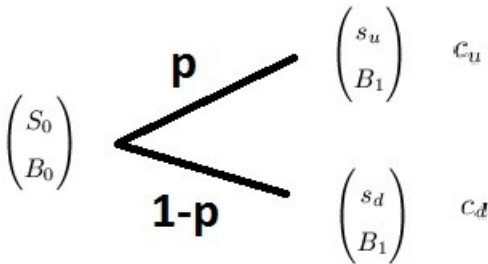
' $\Leftarrow$ ': this follows from the risk-neutral construction derived later in this section (verify!)

What's the price of an option in this market?

- now consider European option with payoff

$$C_1(\omega) = \begin{cases} c_u, & \text{if } \omega = \text{"up"} \\ c_d, & \text{if } \omega = \text{"down"} \end{cases}$$

- for call: $c_u = \max\{s_u - K, 0\}$ and $c_d = \max\{s_d - K, 0\}$
- what is price at $t = 0$ for call?



Method 1: pricing by replication

- construct portfolio V with $V_1(\omega) = C_1(\omega)$ for all ω
 - as portfolio generates exactly the same pay-offs as option does (for all states-of-the-world), the portfolio is called *replicating* portfolio
 - Why? As there are no intermediate cashflows, we must have $C_0 = \phi S_0 + \psi B_0$. Otherwise we would create arbitrage opportunities (this is the *law of one price*). So via the replicating portfolio we find the no-arbitrage price of the option
- notice that we have to determine ϕ and ψ such that

$$c_u = \phi s_u + \psi B_1$$

$$c_d = \phi s_d + \psi B_1$$

which yields

$$\phi = \frac{c_u - c_d}{s_u - s_d} \text{ and } \psi = \frac{c_d s_u - s_d c_u}{B_1(s_u - s_d)}$$

- price of option (at $t = 0$):

$$C_0 = \phi S_0 + \psi B_0$$

Method 2: risk-neutral pricing

Motivation

- people would like to have pricing formula of the form:

$$\frac{A_0}{B_0} = \mathbb{E} \left[\frac{A_1}{B_1} \right]$$

for all assets A , i.e. price equals expected discounted cashflows

- note that

$$\frac{A_0}{B_0} = \mathbb{E} \left[\frac{A_1}{B_1} \right] \iff \mathbb{E} \left[\frac{A_1}{A_0} \right] = \frac{B_1}{B_0}$$

- however, already for $A = S$ this almost never holds true in real financial markets:

$$\mathbb{E} \left[\frac{S_1}{S_0} \right] = \frac{B_1}{B_0},$$

would mean that expected (gross) return on stock is same as return on riskless asset

Intermezzo: a biased coin can save the day

- Suppose we throw a *fair* coin, $\Omega = \{H, T\}$, and you get pay-off $X = 1$ if $\omega = H$ and $X = 0$ if $\omega = T$
- the price of entering this game is 0.3 (given)
- (for some reason) we want to have the identity

$$\text{price} = \mathbb{E}[X] = \text{Prob}(\{H\})$$

- as the coin is fair the equation does not hold true if we use the true, “real-world”, probability measure ($0.3 \neq 0.5$)
- however: we can just create *artificial* probability measure $\mathbb{Q}$ with $\mathbb{Q}(\{H\}) = 0.3$, which does lead to

$$\text{price} = \mathbb{E}_{\mathbb{Q}}[X]$$

Method 2: risk-neutral pricing

- people would like to have pricing formula of the form:

$$\frac{A_0}{B_0} = \mathbb{E} \left[\frac{A_1}{B_1} \right] \iff \mathbb{E} \left[\frac{A_1}{A_0} \right] = \frac{B_1}{B_0} \quad (\star)$$

for all assets A

- if we use “real-world” probabilities then this identity will typically not hold for $A = S$
- could we repair identity by using artificial probability measure $\mathbb{Q}$ instead of “real-world” measure $\mathbb{P}$?
- solving

$$\frac{S_0}{B_0} = \mathbb{E}_{\mathbb{Q}} \left[\frac{S_1}{B_1} \right] = q \frac{s_u}{B_1} + (1 - q) \frac{s_d}{B_1}$$

yields

$$q = \frac{\frac{B_1}{B_0} - \frac{s_d}{S_0}}{\frac{s_u}{S_0} - \frac{s_d}{S_0}}$$

- if $q \in [0, 1]$ then we indeed have found artificial probability measure for which $(\star)$ is true by construction (for $A = B$ the identity is trivial)

Recall earlier theorem: there is no arbitrage if and only if

$$0 < \frac{s_d}{S_0} < \frac{B_1}{B_0} < \frac{s_u}{S_0}$$

which occurs exactly if $q \in (0, 1)$. So we have proved:

Intermediate result

The market is free of arbitrage opportunities if and only if there exists a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that S/B is a martingale under $\mathbb{Q}$, i.e.

$$\frac{S_0}{B_0} = \mathbb{E}_{\mathbb{Q}}\left[\frac{S_1}{B_1}\right].$$

Method 2: risk-neutral pricing

Terminology:

- As S/B is martingale under $\mathbb{Q}$ we have

$$\frac{S_0}{B_0} = \mathbb{E}_{\mathbb{Q}} \left[\frac{S_1}{B_1} \right],$$

which yields

$$\frac{B_1}{B_0} = \mathbb{E}_{\mathbb{Q}} \left[\frac{S_1}{S_0} \right],$$

i.e. the expected (gross) return on the risky asset S , under $\mathbb{Q}$, is equal to the return/interest on the riskless asset B .

- $\mathbb{Q}$ is often called “risk-neutral measure”
 - Please note that $\mathbb{Q}$ is *artificial* measure and does not describe actual probabilities!
- $\mathbb{Q}$ is also called *equivalent martingale measure*

Method 2: risk-neutral pricing

So we have repaired the formula $S_0/B_0 = \mathbb{E}[S_1/B_1]$, but it was quite a discussion. What is the true benefit?

- introduce, as before, an option to the market (with payoff C_1 at $t = 1$)
- we already know that there is a replicating portfolio (ϕ, ψ) such that $V_1(\omega) = \phi S_1(\omega) + \psi B_1 = C_1(\omega)$ for all ω
- and, if there are no arbitrage opportunities, $C_0 = \phi S_0 + \psi B_0$
- now note that

$$\mathbb{E}_{\mathbb{Q}} \left[\frac{C_1}{B_1} \right] = \phi \mathbb{E}_{\mathbb{Q}} \left[\frac{S_1}{B_1} \right] + \psi = \phi \frac{S_0}{B_0} + \psi = \frac{1}{B_0} (\phi S_0 + \psi B_0) = \frac{C_0}{B_0},$$

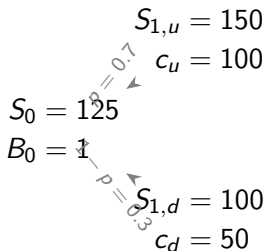
so we can calculate for any option (or asset) its price at $t = 0$ via the above identity as soon as we know $\mathbb{Q}$! There is no need to calculate the replicating portfolios.

Worked example: one-period model

Setup (all numbers reproducible in the accompanying spreadsheet):

- $S_0 = 125$, $B_0 = 1$
- interest: $B_1 = 1.02$ (i.e. $r = 2\%$)
- $S_{1,u} = 150$, $S_{1,d} = 100$
- real-world prob.
 $p = \mathbb{P}(\text{up}) = 0.7$
- claim: call with $K = 50$,
hence

$$c_u = 100, \quad c_d = 50$$



Worked example: pricing by replication

Solve $c_u = \phi s_u + \psi B_1$ and $c_d = \phi s_d + \psi B_1$:

$$\phi = \frac{c_u - c_d}{s_u - s_d} = \frac{100 - 50}{150 - 100} = 1,$$

$$\psi = \frac{c_d s_u - s_d c_u}{B_1(s_u - s_d)} = \frac{50 \cdot 150 - 100 \cdot 100}{1.02 \cdot 50} = -\frac{2500}{51} \approx -49.02.$$

No-arbitrage price of the claim at $t = 0$:

$$C_0 = \phi S_0 + \psi B_0 = 125 - 49.02 = 75.98.$$

Check (the portfolio reproduces the payoff in both states):

$$\phi s_u + \psi B_1 = 150 - 50 = 100 = c_u, \quad \phi s_d + \psi B_1 = 100 - 50 = 50 = c_d.$$

Worked example: risk-neutral pricing

Risk-neutral probability (makes S/B a $\mathbb{Q}$ -martingale):

$$q = \frac{\frac{B_1}{B_0} - \frac{s_d}{S_0}}{\frac{s_u}{S_0} - \frac{s_d}{S_0}} = \frac{1.02 - 0.80}{1.20 - 0.80} = \frac{0.22}{0.40} = 0.55.$$

Price as discounted $\mathbb{Q}$ -expectation of the payoff:

$$C_0 = \frac{1}{B_1} \mathbb{E}_{\mathbb{Q}}[C_1] = \frac{0.55 \cdot 100 + 0.45 \cdot 50}{1.02} = \frac{77.5}{1.02} = 75.98,$$

identical to the replication price — as it must be.

Worked example: why a *different* measure?

The price is *not* the discounted real-world expectation. Under $\mathbb{P}$ ($p = 0.7$):

$$\frac{1}{B_1} \mathbb{E}_{\mathbb{P}}[S_1] = \frac{0.7 \cdot 150 + 0.3 \cdot 100}{1.02} = \frac{135}{1.02} = 132.35 \neq 125 = S_0,$$

$$\frac{1}{B_1} \mathbb{E}_{\mathbb{P}}[C_1] = \frac{0.7 \cdot 100 + 0.3 \cdot 50}{1.02} = \frac{85}{1.02} = 83.33 \neq 75.98.$$

Under the risk-neutral $\mathbb{Q}$ ($q = 0.55$):

$$\frac{1}{B_1} \mathbb{E}_{\mathbb{Q}}[S_1] = \frac{0.55 \cdot 150 + 0.45 \cdot 100}{1.02} = \frac{127.5}{1.02} = 125 = S_0.$$

Only $\mathbb{Q}$ reproduces today's traded prices; this is exactly why we price under $\mathbb{Q}$ rather than $\mathbb{P}$.

Quiz

Take the one-period model with $S_0 = 100$, $B_0 = 1$, $B_1 = 1$ (so $r = 0$), and

$$S_{1,u} = 120, \quad S_{1,d} = 90.$$

- 1 Is the market free of arbitrage?
- 2 Find the risk-neutral probability q of the “up” state.
- 3 Price a call with strike $K = 100$ (payoffs $c_u = 20$, $c_d = 0$).

Method 2: risk-neutral pricing

First fundamental theorem of asset pricing (for the toy model)

The market is free of arbitrage opportunities if and only if there exists a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ such that A/B is a martingale, under $\mathbb{Q}$, for all assets A .

Application We can price options by applying this theorem twice:

- Write down model for B and S (without arbitrage opportunities).
- Apply theorem (with $A = B$ and $A = S$; note that for $A = B$ the 'martingale-property' is trivial). This yields unique $\mathbb{Q}$. ($\star$)
- Now introduce an option to market with payoff C_1 . We want to set price such that we do not create arbitrage opportunity.
- Apply theorem again: A/B should be a martingale for $A = B, S, C$. For $A = S$ we obtain unique $\mathbb{Q}$ from previous step ($\star$). And $A = C$ yields

$$C_0 = B_0 \mathbb{E}_{\mathbb{Q}}[C_1/B_1].$$

Method 2: risk-neutral pricing

Proof

' $\Rightarrow$ ': exercise - check that we have proved this implication (via explicit construction of $\mathbb{Q}$)

' $\Leftarrow$ ':

Consider portfolio V with $V_1 \geq 0$ and $V_1(\omega) > 0$ for at least one state-of-the-world. If we prove that $V_0 > 0$, then we have shown that arbitrage opportunities do not exist.

Note that we have (why?!)

$$\begin{aligned} 0 < \mathbb{E}_{\mathbb{Q}} \left[\frac{V_1}{B_1} \right] &= \phi \mathbb{E}_{\mathbb{Q}} \left[\frac{S_1}{B_1} \right] + \psi \\ &= \phi \frac{S_0}{B_0} + \psi = \frac{1}{B_0} (\phi S_0 + \psi B_0) = \frac{V_0}{B_0}, \end{aligned}$$

which yields $V_0 > 0$.

Method 3: the pricing kernel

A third, equivalent route. A **pricing kernel** (a.k.a. stochastic discount factor) is a random variable $m > 0$ such that, for every asset A ,

$$A_0 = \mathbb{E}_{\mathbb{P}}[m A_1].$$

In the toy model m takes two values m_u, m_d , determined from the basic assets:

$$B_0 = \mathbb{E}_{\mathbb{P}}[m B_1] = B_1(p m_u + (1 - p)m_d),$$

$$S_0 = \mathbb{E}_{\mathbb{P}}[m S_1] = p m_u s_u + (1 - p)m_d s_d.$$

Two equations, two unknowns. Then *any* claim is priced by

$$C_0 = \mathbb{E}_{\mathbb{P}}[m C_1] = p m_u c_u + (1 - p)m_d c_d.$$

Method 3: relation to $\mathbb{Q}$, and the worked example

The three methods are one and the same. Solving the two equations gives

$$m_\omega = \frac{B_0}{B_1} \frac{q_\omega}{p_\omega}, \quad \text{i.e.} \quad m = \underbrace{\frac{B_0}{B_1}}_{\text{discount}} \times \underbrace{\frac{d\mathbb{Q}}{d\mathbb{P}}}_{\text{change of measure}}.$$

For our numbers ($p = 0.7$, $q = 0.55$, $B_1 = 1.02$):

$$m_u = \frac{0.55}{1.02 \cdot 0.7} \approx 0.770, \quad m_d = \frac{0.45}{1.02 \cdot 0.3} \approx 1.471,$$

$$C_0 = 0.7 \cdot 0.770 \cdot 100 + 0.3 \cdot 1.471 \cdot 50 \approx 75.98,$$

again the same price. Note $m_d > m_u$: a payoff in the “down” (low- S) state is more expensive — the kernel encodes the price of risk.

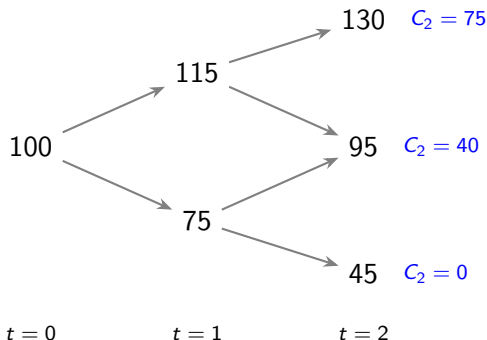
Multi-period binomial model

The one-period idea extends directly to several trading dates.

Consider a *two-period*, recombining model ($t = 0, 1, 2$): we may trade and rebalance at $t = 0$ and $t = 1$.

Bond: $B_0 = 1$, $B_1 = 1.05$,
 $B_2 = 1.11$.

Claim: call with $K = 55$, so the
payoff $C_2 = \max\{S_2 - 55, 0\}$
equals 75, 40, 0 in the three
terminal states.



Pricing by backward induction

Key observation: each *one-period sub-tree* is exactly the model we already solved.

- start at maturity: the values at $t = 2$ are the known payoffs;
- step back one period at a time. At a node at time t with one-period successors “ u ” / “ d ”, the no-arbitrage value is

$$V_t = \frac{B_t}{B_{t+1}} \left(q V_{t+1}^u + (1-q) V_{t+1}^d \right), \quad q = \frac{B_{t+1}/B_t - s_d/s_t}{s_u/s_t - s_d/s_t},$$

equivalently, replicate that single period with (ϕ, ψ) ;

- the resulting strategy is *self-financing*: at $t = 1$ we rebalance, and the value of the old position exactly funds the new one (no cash injected/withdrawn).

We illustrate with replication (the spreadsheet also verifies the risk-neutral route).

Two-period example: step back to $t = 1$

At each $t = 1$ node we solve a one-period problem for the $t = 2$ payoffs.

Up node ($S_{1,u} = 115 \rightarrow 130/95$, payoffs 75/40):

$$\phi_{1,u} = \frac{75 - 40}{130 - 95} = 1, \quad \psi_{1,u} \approx -49.55, \quad V_{1,u} \approx 62.97.$$

Down node ($S_{1,d} = 75 \rightarrow 95/45$, payoffs 40/0):

$$\phi_{1,d} = \frac{40 - 0}{95 - 45} = 0.8, \quad \psi_{1,d} \approx -32.43, \quad V_{1,d} \approx 25.95.$$

Two-period example: step back to $t = 0$

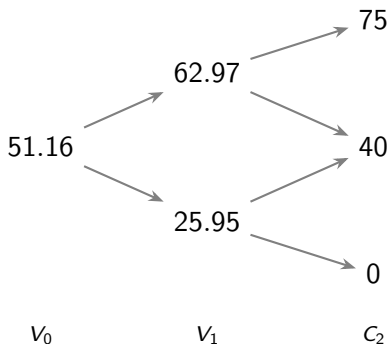
Now treat $V_{1,u} \approx 62.97$ and $V_{1,d} \approx 25.95$ as the “payoffs” of a one-period problem at $t = 0$:

$$\phi_0 = \frac{62.97 - 25.95}{115 - 75} \approx 0.926, \quad \psi_0 \approx -41.41,$$

$$V_0 = \phi_0 S_0 + \psi_0 B_0 = 0.926 \cdot 100 - 41.41 \approx 51.16.$$

So the no-arbitrage price of the two-period call is ≈ 51.16 .

At $t = 1$ we *rebalance* from (ϕ_0, ψ_0) to $(\phi_{1,\cdot}, \psi_{1,\cdot})$; self-financing means the rebalancing costs nothing.



Food for thought (for the curious / advanced student)

Suppose the stock can move to *three* values at $t = 1$ (“up”, “middle”, “down”) instead of two, with the bond unchanged.

- Can you still replicate an arbitrary payoff (c_u, c_m, c_d) using only the stock and the bond?
- How many probability measures $\mathbb{Q}$ make S/B a martingale now?
- What does this imply for the *uniqueness* of the no-arbitrage price?

This is the essence of *market incompleteness*: with more states than independent traded assets, replication fails and $\mathbb{Q}$ is no longer unique. We return to completeness in Part 3.

For the toy model we have seen that we can price options by

- constructing a replicating portfolio,
- using risk-neutral pricing, and
- using the pricing kernel.

All three give the *same* price, and they extend to the multi-period tree via backward induction.

For the continuous-time case we will see that exactly the same ideas are used. Unfortunately (?), the math gets a bit more difficult.